

DSMC Calculator — Mathematical and Physical Formulation

Technical Documentation — Version 1.0

1. Introduction

The **DSMC Calculator** is an engineering calculation tool for preliminary design and numerical setup of Direct Simulation Monte Carlo (DSMC) simulations for rarefied and hypersonic gas flows.

The calculator evaluates thermodynamic, molecular, transport, dimensionless-flow, and preliminary DSMC discretization quantities, including pressure, number density, density, speed of sound, Mach number, VHS collision cross-section, mean free path, mean thermal speed, collision time, viscosity, Knudsen number, Reynolds number, cell size, particle weighting, time step, recovery temperature, Eckert reference temperature, and flat-plate boundary-layer estimates.

It also provides inverse calculations for: 1. Required number density and pressure for a prescribed Knudsen number. 2. Required velocity for a prescribed Reynolds number.

2. Fundamental Physical Constants

The Boltzmann constant is

$$[k_B = 1.380649 \times 10^{-23} \text{ J/K}.]$$

The molecular mass (m) is expressed in kg.

The default molecular parameters in the calculator correspond approximately to argon:

$$[m = 66.3 \times 10^{-27} \text{ kg},]$$

$$[d_{\text{ref}} = 4.17 \times 10^{-10} \text{ m},]$$

$$[\sigma = 0.81,]$$

$$[\gamma = 1.667,]$$

$$[Pr = 0.667.]$$

3. Ideal-Gas Relations

3.1 Ideal-gas equation

The gas is treated as an ideal gas:

$$\left[P = nk_B T \right]$$

where (P) is pressure [Pa], (n) is molecular number density [m^{-3}], (k_B) is the Boltzmann constant, and (T) is temperature [K].

Therefore,

$$\left[n = \frac{P}{k_B T} \right]$$

and

$$\left[P = nk_B T \right].$$

3.2 Mass density

The macroscopic mass density is

$$\left[\rho = nm \right]$$

where () is [kg/m^3].

3.3 Specific gas constant

$$\left[R = \frac{k_B}{m} \right]$$

with units J/(kg K).

4. Speed of Sound and Mach Number

For an ideal gas, the speed of sound is

$$\left[a = \sqrt{\gamma RT} \right]$$

or equivalently

$$\left[a = \sqrt{\gamma \frac{k_B}{m} T} \right].$$

The Mach number is

$$\left[M = \frac{U}{a} \right]$$

and therefore

$$\left[U = Ma \right].$$

Here (U) is the bulk-flow velocity [m/s].

5. Variable Hard Sphere (VHS) Model

The calculator uses the Variable Hard Sphere (VHS) model to represent molecular collision and viscosity behavior.

The principal VHS parameters are the reference diameter (d_{ref}), reference temperature (T_{ref}), and viscosity index (ω).

5.1 VHS collision cross-section

The total collision cross-section is

$$\left[\sigma_T = \pi d_{\text{ref}}^2 \left(\frac{T_{\text{ref}}}{T} \right)^{\omega-1/2} \right]$$

where (σ_T) is [m^2].

At ($T=T_{\text{ref}}$),

$$\left[\sigma_T = d_{\text{ref}}^2 \right]$$

6. Mean Free Path

The molecular mean free path is calculated from

$$\left[\lambda = \frac{1}{\sqrt{2} n \sigma_T} \right]$$

where (λ) is [m].

Thus, for fixed molecular properties and temperature,

$$\left[\lambda \propto \frac{1}{n} \right]$$

Increasing number density therefore decreases the mean free path.

7. Mean Thermal Molecular Speed

The mean thermal molecular speed is

$$\left[\bar{c} = \sqrt{\frac{8k_B T}{\pi m}} \right]$$

Consequently,

$$\left[\bar{c} \propto \sqrt{\frac{T}{m}} \right]$$

The thermal speed increases with temperature and decreases with molecular mass.

8. Mean Collision Time

The characteristic mean collision time is

$$\left[\tau = \frac{\lambda}{\bar{c}} \right]$$

or

$$\left[\tau = \frac{1}{\sqrt{2} n \sigma_T \bar{c}} \right]$$

The collision time is a characteristic molecular time scale.

9. VHS Dynamic Viscosity

The reference viscosity is calculated using

$$\left[\mu_{\text{ref}} = \frac{15 \sqrt{\pi m k_B T_{\text{ref}}}}{2 \pi d_{\text{ref}}^2 (5 - 2\omega)(7 - 2\omega)} \right]$$

and the viscosity at the actual gas temperature is

$$\left[\mu = \mu_{\text{ref}} \left(\frac{T}{T_{\text{ref}}} \right)^\omega \right]$$

Here () is dynamic viscosity [Pa · s].

10. Knudsen Number

The Knudsen number is

$$\left[Kn = \frac{\lambda}{L_{\text{Kn}}} \right]$$

where (L_{Kn}) is the selected characteristic length [m].

The calculator uses the following engineering classification:

Knudsen number	Flow regime
(Kn<0.01)	Continuum
(0.01<Kn<0.1)	Slip flow

Knudsen number	Flow regime
(0.1 < Kn < 10)	Transitional
(Kn > 10)	Free molecular

These limits are engineering guidance rather than strict physical discontinuities.

11. Solving for Required Density at a Target Knudsen Number

For a specified target Knudsen number,

$$[\text{Kn}_{\text{target}} = \frac{\lambda_{\text{target}}}{L_{\text{Kn}}},]$$

so

$$[\lambda_{\text{target}} = K n_{\text{target}} L_{\text{Kn}}.]$$

Starting from

$$[= \frac{1}{\sqrt{2} n \sigma_T},]$$

the required number density is

$$[n_{\text{req}} = \frac{1}{\sqrt{2} \sigma_T \lambda_{\text{target}}}]$$

or

$$[n_{\text{req}} = \frac{1}{\sqrt{2} \sigma_T K n_{\text{target}} L_{\text{Kn}}}.]$$

The corresponding pressure is

$$[P_{\text{req}} = n_{\text{req}} k_B T]$$

and the corresponding mass density is

$$[\rho_{\text{req}} = n_{\text{req}} m.]$$

The associated collision time is

$$[\tau_{\text{target}} = \frac{\lambda_{\text{target}}}{\bar{c}}.]$$

12. Reynolds Number

The Reynolds number is

$$\left[Re = \frac{\rho U L_{Re}}{\mu} \right]$$

where: - (Re) is dimensionless, - (ρ) is density [kg/m^3], - (U) is velocity [m/s], - (L_{Re}) is the characteristic Reynolds-number length [m], - (μ) is dynamic viscosity [$\text{Pa} \cdot \text{s}$].

Using ($=\text{nm}$),

$$\left[Re = \frac{nm U L_{Re}}{\mu} \right]$$

The Reynolds number measures the relative importance of inertial and viscous effects.

13. Solving for Required Velocity at a Target Reynolds Number

Starting with

$$\left[Re = \frac{\rho U L_{Re}}{\mu} \right],$$

the required velocity is

$$\left[U_{req} = \frac{Re_{target} \mu}{\rho L_{Re}} \right]$$

The corresponding Mach number is

$$\left[M_{req} = \frac{U_{req}}{a} \right]$$

Thus, a specified Reynolds number can be translated into the required flow velocity and Mach number for the selected thermodynamic and molecular state.

14. DSMC Computational Cell Size

For a cubic-equivalent cell, the characteristic cell dimension is

$$\left[\Delta x = V_{cell}^{1/3} \right]$$

where (V_{cell}) is cell volume [m^3].

If the domain volume and number of cells are specified,

$$\left[V_{\text{cell}} = \frac{V_{\text{domain}}}{N_{\text{cells}}} \right]$$

and therefore

$$\left[\Delta x = \left(\frac{V_{\text{domain}}}{N_{\text{cells}}} \right)^{1/3} \right]$$

15. Number of Real Molecules per Cell

The number of physical molecules contained in a cell is

$$\left[N_{\text{real}} = nV_{\text{cell}} \right]$$

This is generally much larger than the number of computational particles used in a DSMC simulation.

16. Simulation Particle Weight

If (N_{ppc}) computational particles are desired per cell, the number of real molecules represented by one computational particle is

$$\left[N_{\text{eq}} = \frac{nV_{\text{cell}}}{N_{\text{ppc}}} \right]$$

This quantity is the molecular statistical weight of one simulation particle.

17. DSMC Time-Step Criteria

The calculator considers two characteristic time-step restrictions.

17.1 Collision-time criterion

$$\left[\Delta t_{\text{coll}} = f_1 \tau \right]$$

where (f_1) is a user-selected safety factor.

The default value is

$$\left[f_1 = 0.2 \right]$$

17.2 Cell-transit criterion

The transport/transit restriction is

$$\left[\Delta t_{\text{trans}} = f_2 \frac{\Delta x}{U + \bar{c}} \right]$$

where (f_2) is a safety factor.

The default value is

$$[f_2 = 0.3.]$$

17.3 Recommended time step

The final time step is

$$\left[\Delta t = \min(\Delta t_{\text{coll}}, \Delta t_{\text{trans}}) \right]$$

or

$$\left[\Delta t = \min \left[f_1 \tau, f_2 \frac{\Delta x}{U + \bar{c}} \right] \right]$$

18. Rarefaction Coefficient

The calculator includes a rarefaction/compression coefficient (C_r).

When pressure is the primary input,

$$\left[P = C_r P_{\text{input}} \right]$$

followed by

$$\left[n = \frac{P}{k_B T} \right]$$

When number density is the primary input,

$$\left[n = C_r n_{\text{input}} \right]$$

followed by

$$\left[P = n k_B T \right]$$

For constant temperature and molecular properties,

$$\left[\frac{1}{n} \right]$$

and hence

$$\left[Kn \propto \frac{1}{n} \right]$$

Therefore, decreasing density increases rarefaction.

19. Recovery Temperature

The calculator uses the following recovery-temperature relation:

$$\left[T_r = T \left[1 + \sqrt{Pr} \frac{\gamma - 1}{2} M^2 \right] \right]$$

where: - (T_r) is recovery temperature [K], - (T) is free-stream temperature [K], - (Pr) is the Prandtl number, - (γ) is the specific heat ratio, - (M) is Mach number.

The corresponding recovery factor used by the calculator is

$$\left[r = \sqrt{Pr} \right]$$

For comparison, the ideal adiabatic stagnation temperature is

$$\left[T_0 = T \left[1 + \frac{\gamma - 1}{2} M^2 \right] \right]$$

The recovery temperature is therefore distinct from the stagnation temperature.

20. Eckert Reference Temperature

The calculator uses the following Eckert-type reference-temperature relation:

$$\left[T^* = T \left[1 + 0.032 M^2 + 0.58 \left(\frac{T_w}{T} - 1 \right) \right] \right]$$

where: - (T^*) is reference temperature [K], - (T_w) is wall temperature [K], - (T) is free-stream temperature [K].

This reference temperature is subsequently used to evaluate reference viscosity and density for the boundary-layer estimate.

21. Reference Viscosity and Density

At the Eckert reference temperature,

$$\left[\mu^* = \mu_{\text{ref}} \left(\frac{T^*}{T_{\text{ref}}} \right)^\omega \right]$$

and

$$\left[\rho^* = \frac{P}{RT^*} \right]$$

These quantities are then used to calculate the local Reynolds number.

22. Local Reynolds Number

The local Reynolds number based on distance (x) from the leading edge is

$$\left[Re_x = \frac{\rho^* U x}{\mu^*} \right]$$

Here (x) is measured from the leading edge and has units of meters.

23. Flat-Plate Boundary-Layer Estimates

The calculator provides classical engineering estimates for a laminar flat-plate boundary layer.

23.1 Boundary-layer thickness

$$\left[\delta = \frac{5x}{\sqrt{Re_x}} \right]$$

where () is the estimated 99%-velocity boundary-layer thickness.

23.2 Momentum thickness

$$\left[\theta = \frac{0.664x}{\sqrt{Re_x}} \right]$$

23.3 Displacement thickness

$$\left[\delta^* = \frac{1.721x}{\sqrt{Re_x}} \right]$$

These are engineering correlations and should not automatically be treated as accurate predictions for separated, strongly rarefied, transitional, reacting, or thermochemical-nonequilibrium flows.

24. Local Knudsen Number

The calculator evaluates a local Knudsen number based on the distance from the leading edge:

$$\left[Kn_x = \frac{\lambda}{x} \right]$$

A warning is generated when

$$\left[Kn_x > 0.1 \right]$$

This indicates that conventional continuum boundary-layer assumptions may become questionable and that rarefaction effects should be considered.

25. Principal Parameter Relationships

The main physical chain used by the calculator can be summarized as

$$\left[P, T \rightarrow n \rightarrow \right]$$

and

$$\left[n, T, d_{\{ref\}}, T_{\{ref\}} \rightarrow T \rightarrow Kn \right]$$

The flow velocity and viscosity then provide

$$\left[\mu, U, L_{\{Re\}} \rightarrow Re \right]$$

For DSMC discretization,

$$\left[V_{\{cell\}}, n \rightarrow N_{\{real\}} \right]$$

and

$$\left[N_{\{real\}}, N_{\{ppc\}} \rightarrow N_{\{eq\}} \right]$$

Finally,

$$\left[\mu, x, U, \bar{c} \rightarrow \Delta t \right]$$

26. Relationship Between Pressure and Knudsen Number

At fixed (T), molecular properties, and characteristic length,

$$\left[n = \frac{P}{k_B T} \right]$$

and

$$\left[\lambda = \frac{1}{\sqrt{2}n\sigma_T} \right]$$

Therefore,

$$\left[\lambda = \frac{k_B T}{\sqrt{2}P\sigma_T} \right]$$

Consequently,

$$\left[Kn = \frac{k_B T}{\sqrt{2}P\sigma_T L_{Kn}} \right]$$

and therefore

$$\left[Kn \propto \frac{1}{P} \right]$$

when the remaining parameters are constant.

This explains why lowering the pressure increases the degree of rarefaction.

27. Relationship Between Reynolds Number and Velocity

From

$$\left[Re = \frac{\rho U L_{Re}}{\mu} \right]$$

if density, viscosity, and characteristic length are fixed,

$$\left[Re \propto U \right]$$

Therefore,

$$\left[U_{req} = \frac{Re_{target}\mu}{\rho L_{Re}} \right]$$

Increasing velocity increases Reynolds number under otherwise identical conditions.

28. DSMC Resolution Indicators

Two important length scales are

$$\left[\lambda \right]$$

and

$$\left[x \right]$$

Their ratio,

$$\left[\frac{\Delta x}{\lambda}, \right]$$

is a useful preliminary DSMC resolution indicator.

If the cell dimension becomes larger than the molecular mean free path,

$$[\text{ x} > ,]$$

the calculator issues a warning.

The appropriate cell-size criterion ultimately depends on the DSMC formulation, collision model, and flow physics; therefore this warning should be treated as a preliminary screening criterion rather than a substitute for numerical verification.

29. Summary of Implemented Equations

Thermodynamic relations

$$[P = nk_{\text{BT}}]$$

$$[\text{ } = nm]$$

$$[R = \frac{k_B}{m}]$$

$$[a = \sqrt{\gamma RT}]$$

$$[M = \frac{U}{a}]$$

VHS model

$$[T = d_{\{\text{ref}\}}^2 \left(\frac{T_{\text{ref}}}{T} \right)^{\omega - 1/2}]$$

$$[\text{ } = \frac{1}{\sqrt{2n\sigma_T}}]$$

$$[\bar{c} = \sqrt{\frac{8k_B T}{\pi m}}]$$

$$[\text{ } = \frac{\lambda}{\bar{c}}]$$

$$[\text{ }_{\{\text{ref}\}} = \frac{15\sqrt{\pi m k_B T_{\text{ref}}}}{2\pi d_{\text{ref}}^2 (5 - 2\omega)(7 - 2\omega)}]$$

$$[\text{ } = \text{ }_{\{\text{ref}\}} \left(\frac{T}{T_{\text{ref}}} \right)^{\omega}]$$

Dimensionless parameters

$$[\text{Kn} = \frac{\lambda}{L_{\text{Kn}}}]$$

$$[\text{Re} = \frac{\rho U L_{\text{Re}}}{\mu}]$$

DSMC discretization

$$[V_{-}\{\text{cell}\} = \frac{V_{\text{domain}}}{N_{\text{cells}}}]$$

$$[x = V_{-}\{\text{cell}\}^{\wedge}\{1/3\}]$$

$$[N_{-}\{\text{real}\} = n V_{-}\{\text{cell}\}]$$

$$[N_{-}\{\text{eq}\} = \frac{n V_{\text{cell}}}{N_{\text{ppc}}}]$$

Time step

$$[t_{-}\{\text{coll}\} = f_1]$$

$$[t_{-}\{\text{trans}\} = f_2 \frac{\Delta x}{U + \bar{c}}]$$

$$[t = \min(\Delta t_{-}\{\text{coll}\}, t_{-}\{\text{trans}\})]$$

Boundary-layer relations

$$[T_{-}r = T \left[1 + \sqrt{Pr} \frac{\gamma-1}{2} M^2 \right]]$$

$$[T^{\wedge}* = T \left[1 + 0.032 M^2 + 0.58 \left(\frac{T_w}{T} - 1 \right) \right]]$$

$$[\wedge* = _{-}\{\text{ref}\} \left(\frac{T^*}{T_{\text{ref}}} \right)^{\wedge}\omega]$$

$$[* = \frac{P}{RT^*}]$$

$$[Re_{-}x = \frac{\rho^* U x}{\mu^*}]$$

$$[= \frac{5x}{\sqrt{Re_x}}]$$

$$[= \frac{0.664x}{\sqrt{Re_x}}]$$

$$[\wedge* = \frac{1.721x}{\sqrt{Re_x}}]$$

$$[Kn_{-}x = \frac{\lambda}{x}]$$

30. Recommended Engineering Workflow

For a new DSMC case:

1. Define the gas molecular properties: [m, d_{-}\{\text{ref}\}, , T_{-}\{\text{ref}\}, , Pr.]
2. Define the thermodynamic state: [T, P {or} n.]
3. Define the flow velocity or Mach number: [U {or} M.]
4. Calculate: [_T, , \bar{c}, \tau, \mu.]
4. Determine the flow regime using: [Kn = \frac{\lambda}{L_{Kn}}.]

4. Determine inertial/viscous scaling using: [$Re = \frac{\rho U L_{Re}}{\mu}$.]
 4. Define the DSMC computational resolution: [$V_{\{cell\}}$, x , $N_{\{ppc\}}$.]
 5. Determine the particle statistical weight: [$N_{\{eq\}}$.]
 6. Determine the preliminary time step: [t .]
 7. For boundary-layer applications, evaluate: [T_r , T^* , Re_x , γ , θ^* , Kn_x .]
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31. Modeling Limitations

The calculator is intended for **preliminary engineering calculations and DSMC setup guidance**.

The results rely on the assumptions and correlations described above, including:

- ideal-gas thermodynamics,
- the selected VHS molecular model,
- the implemented mean-free-path expression,
- the specified transport-property correlations,
- classical flat-plate boundary-layer correlations.

The calculator does not by itself replace detailed DSMC verification, grid and time-step sensitivity studies, particle-number convergence studies, or more advanced thermochemical and molecular models where those are required.

In particular, the flat-plate boundary-layer estimates should be used cautiously for strongly rarefied, transitional, separated, chemically reacting, or thermochemical-nonequilibrium flows.

32. Final Remarks

The DSMC Calculator connects macroscopic flow conditions with molecular and numerical scales required for preliminary DSMC planning.

The principal inverse calculations can be summarized as

$$\left[Kn_{\text{target}} \rightarrow \lambda_{\text{target}} \rightarrow n_{\text{req}} \rightarrow P_{\text{req}} \right]$$

and

$$\left[Re_{\text{target}} \rightarrow U_{\text{req}} \rightarrow M_{\text{req}} \right]$$

These relationships allow the user to determine operating conditions corresponding to specified rarefaction and Reynolds-number requirements before performing a DSMC simulation.